

GWANYEOP KIM

Undergraduate Student

☎ 010-7152-1573

✉ kgy020708@naver.com

📍 Konkuk University, Seoul

RESEARCH INTERESTS

My primary research goal is to improve the reliability and reasoning efficiency of large language models through structured agent architectures and reinforcement learning.

I am particularly interested in agentic AI, Multi-Agent System, retrieval-augmented generation, reinforcement learning for structured reasoning, and BioNLP.

EDUCATION

Konkuk University | Mar.2025 - Present

Dept. Computer Science & Engineering

GPA : 3.68 / 4.5

Republic of Korea Air Force | Mar.2023 - Dec.2024

Military Service

Information System Management Specialist

Sejong University | Mar.2021 - Dec.2022

Dept. Computer Science & Engineering

GPA : 3.85 / 4.5

RESEARCH EXPERIENCE

Data Mining and Information Systems Lab

(DMIS Lab), Korea University | March.2026 - Present

Undergraduate Research Intern

- Research Area: Agentic AI, Multi Agent System, BioNLP

Natural Language Processing Lab (NLP Lab), UNIST | May.2025 - Feb.2026

Undergraduate Research Intern

- Research Area: RAG Optimization, Efficient Reasoning, Agentic LLM Systems
 - Conducted research on improving RAG systems through adaptive optimization strategies
 - Designed and implemented reward-based retrieval-generation balancing mechanisms
 - Led the development of a research prototype resulting in a first-author paper under review
-

PUBLICATIONS

Submission

"AMORO-RAG : Adaptive Multi-Objective Reinforcement Optimization for RAG"

SIGIR 2026 Under Review

- 1st author
 - Proposed a dynamic RAG optimization framework that improves retrieval quality and reasoning reliability through query-specific reward weighting
 - Designed query-specific reward weighting mechanism
-

AWARDS

Hosted/Organized by : Ministry of Trade, Industry and Energy / Korea Robot Industry Advancement Institute, Korea Robot Industry Association

2021 16th IRC AI Robot Autonomous Mission Competition

- in the Autonomous Flight Category
 - Gold Prize (1st Place)
-

SKILLS AND CERTIFICATIONS

- Language : English (Intermediate, TOEIC : 830), Korean (Native)
 - Programming Languages : Python (Advanced), C (Intermediate)
 - Data Science & Machine Learning : Deep Learning/NLP (Advanced), Machine Learning (Advanced), Computer Vision (Basic)
 - Frameworks and Libraries : PyTorch/HuggingFace/FAISS (Advanced), TensorFlow/LangChain/NumPy/Pandas (Intermediate)
 - Certificates : SQLD, Information Processing Technician
-

PROJECTS

Konkuk University CSE graduation project | Feb.2026 - Present

Agentic RAG-based Medication Recommendation System

- Developed an agentic RAG-based system for symptom-driven medication recommendation, where the model performs iterative retrieval and analysis to infer appropriate over-the-counter (OTC) drugs.
 - Designed a structured decision pipeline that constrains recommendations to safe, non-prescription medications, improving reliability and safety in domain-specific generation.
-

ACTIVITIES

- LG Aimers 7th Cohort | Jul.2025 - Aug.2025
- 2025 Dacon Food and Beverage Menu Demand Prediction AI Online Hackathon, Top 10%